Imagine the following scenario: you work at a phone manufacturing company in the quality control department, and your current project is to stress-test the robustness of the casing of the company's newest product. Specifically, your team wants to find out from which floor of a 10-story building one can drop the phone to the ground and not break it. A few rules apply:

- The engineers who made the phone are sure that it won't break if dropped from the 1st floor.
- If the phone breaks when dropped from a given floor, then it will also break when dropped from a higher floor.

You are tasked with finding the highest floor that one can drop the phone from and it still won't break—we will denote this unknown floor as $X$—by conducting trials. That is, you will drop actual copies of the phone from specific floors in order to determine $X$. The question is, how should you choose which floors to drop the phones from, in order to find $X$? Since the phones are expensive, you need to conduct as few trials as possible and would like to leverage information theory to aid the search.

1. Assume that by taking into account physics and the makings of the phone, the engineers have an initial guess as to which floor might be possible.
   Specifically, the prior distribution of $X$ is exponential in that the probability that $X$ is equal to a number is inversely exponential with respect to that number: $Pr(X = n) = 1 / 2^n$, for $n = 1, 2, \ldots, 9$; the probability corresponding to the highest, 10th floor is $Pr(X = 10) = 1 / 2^9$. So the probability that $X = 1$ is 50%, and this probability is cut in half as the number increases.

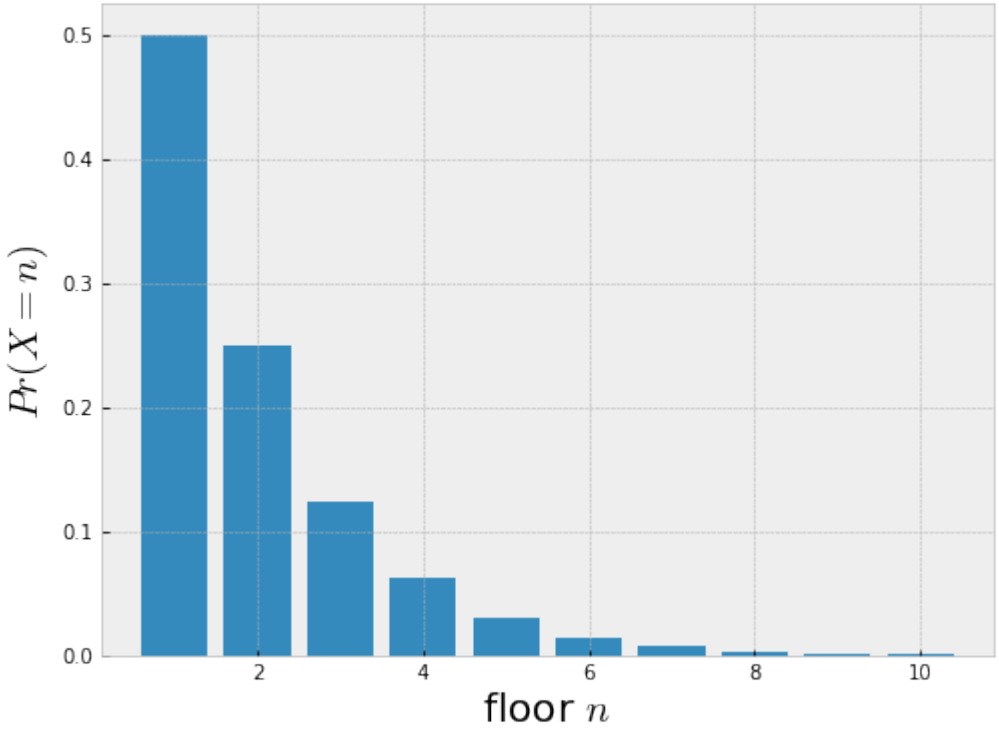



**Figure 6.14 Probability that X is equal to a number between 1 and 10 (that is, the probability that a floo is the highest floor that doesn't cause the phone to break when dropped).**

©Manning Publications Co. To comment go to liveBook
https://livebook.manning.com/#!/book/bayesian-optimization-in-action/discussion

**Licensed to ashwin sampath <ashwin.manning@gmail.com>**